Department of Electrical and Computer Engineering

North South University

Senior Design Project

Speech-to-Visual Storytelling:

Context Segmentation and RAG-Enhanced AI-Generated Imagery for Enhanced Video Comprehension

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Summer, 2024

LETTER OF TRANSMITTAL

April, 2025

To

Dr. Mohammad Abdul Matin

Chairman,

Department of Electrical and Computer Engineering

North South University, Dhaka

Subject: **Submission of Capstone Project Report on "Explainable AI-driven Phishing Website Detection Using Transfer Learning."**

Dear Sir,

With due respect, we would like to submit our **Capstone Project Report** on **"Explainable AI-driven Phishing Website Detection Using Transfer Learning."** as a part of our BSc program. The report deals with a multi-level phishing website detection system. This project was very valuable to us as it helped us gain experience in the cyber security field. We tried to the maximum competence to meet all the dimensions required from this report.

We will be highly obliged if you receive this report and provide your valuable judgment. It is our immense pleasure that you find this report helpful and informative and have an apparent perspective.

Sincerely Yours,

.....

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APPROVAL

Redwan Ali Rafi, Talha Taiyoub, Nafsi Omar Prokrety, and Madiha Jarrin from the Electrical and Computer Engineering Department of North South University have worked on the Senior Design Project titled “Explainable AI-driven Phishing Website Detection Using Transfer Learning.” under the supervision of Dr. Riasat Khan partial fulfillment of the requirement for the degree of Bachelors of Science in Engineering and has been accepted as satisfactory.

Supervisor’s Signature

…….

Dr. Mohammad Ashrafuzzaman Khan

Associate Professor

Department of Electrical and Computer Engineering

North South University

Dhaka, Bangladesh.

Chairman’s Signature

…….

Dr. Mohammad Abdul Matin

Professor

Department of Electrical and Computer Engineering

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DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

Students' names & Signatures

1. Talha Taiyoub

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ACKNOWLEDGEMENTS

The authors would like to express their heartfelt gratitude towards their project and research supervisor, Dr. Mohammad Ashrafuzzaman Khan, Associate Professor, Department of Electrical and Computer Engineering, North South University, Bangladesh, for his invaluable support, precise guidance and advice pertaining to the experiments, research and theoretical studies carried out during the course of the current project and also in the preparation of the current report.

Furthermore, the authors would like to thank the Department of Electrical and Computer Engineering, North South University, Bangladesh, for facilitating the research. We would also like to thank my friends for helping us with this project. The authors would also like to thank their loved ones for their countless sacrifices and continual support.

ABSTRACT

Speech-to-Visual Storytelling: Context Segmentation and RAG-Enhanced AI-Generated Imagery for Enhanced Video Comprehension

This paper presents a novel framework for Speech-to-Visual Storytelling, integrating context segmentation and Retrieval-Augmented Generation (RAG) to produce coherent and contextually enriched AI-generated imagery for enhanced video comprehension. Leveraging automatic speech recognition (ASR), the system segments spoken narratives into meaningful units, which are then used to query external knowledge bases and enrich the generative process. The proposed approach combines natural language processing and text-to-image generation models, enhanced by RAG mechanisms, to produce visuals that align closely with narrative context and intent. Experimental evaluations demonstrate the effectiveness of our method in generating semantically relevant and temporally coherent imagery, significantly improving user engagement and understanding in educational and media content applications. This work lays the foundation for more immersive and intelligent multimedia experiences driven by multimodal AI.

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Chapter 1 Introduction

1.1 Background and Motivation

With the exponential growth of multimedia content across digital platforms, understanding and engaging with spoken content in videos has become a crucial challenge. Traditional video comprehension relies heavily on subtitles or manual annotations, which often fail to capture the full narrative or emotional tone. In parallel, the advancement of artificial intelligence in natural language processing (NLP) and computer vision has opened up new possibilities for enhancing comprehension and accessibility.

One promising approach is speech-to-visual storytelling—a technique that translates spoken words into dynamic, AI-generated visual scenes. This project further enhances the technique by integrating context segmentation and Retrieval-Augmented Generation (RAG), enabling more coherent and context-aware imagery generation. The goal is to create a seamless experience where spoken content is automatically interpreted, segmented by context, and illustrated through accurate and expressive visuals.

Challenges Addressed:

Loss of Engagement and Context: Conventional educational and informational videos often struggle to maintain viewer engagement or provide sufficient context for deeper understanding.

Accessibility Gaps: Hearing-impaired users or individuals with language barriers often face difficulty in comprehending spoken content, especially in non-subtitled or technical videos.

Manual Content Creation Bottlenecks: Producing visual content from speech typically requires manual effort, which is time-consuming and resource-intensive.

AI Interpretation Limitations: Existing AI systems often generate generic or out-of-context imagery due to a lack of contextual understanding and integration with relevant background knowledge.

Motivation:

The motivation for this project lies in bridging the gap between auditory content and visual comprehension. By transforming speech into segmented, visually coherent narratives, we can enhance understanding, retention, and accessibility. The incorporation of RAG enables the system to retrieve relevant information from a knowledge base, ensuring that generated visuals are not only accurate but also semantically rich.

This innovation is especially beneficial for educational platforms, content creators, mental health applications, and assistive technologies, where the fusion of speech and visuals can amplify impact and usability.

1.2 Purpose and Goal of the Project

The primary purpose of this project is to develop a system that can convert spoken content into meaningful and context-aware visual storytelling using cutting-edge AI techniques. The system utilizes context segmentation to break down long-form speech into thematic units and then applies RAG-based generation to synthesize visuals that accurately reflect each segment's context.

Core Goals: To create a pipeline that converts real-time or pre-recorded speech into semantically segmented narratives.

To leverage Retrieval-Augmented Generation (RAG) for generating high-quality, knowledge-enriched imagery that complements spoken content.

To improve accessibility and engagement in video content through automated visual enhancement.

To explore the integration of large language models and image generation frameworks in a unified storytelling system.

Benefits of the Approach:

Enhanced Comprehension: Visual aids generated based on context help viewers better understand complex or abstract spoken content.

Inclusive Content Consumption: Individuals with hearing or cognitive challenges gain improved access to multimedia through visual storytelling.

Efficiency and Scalability: Reduces the need for human labor in visual production, making it scalable for educators, influencers, and institutions.

Knowledge-Rich Visuals: RAG ensures visuals are not only coherent but enriched with retrieved factual data, increasing authenticity and relevance. Ultimately, this project aims to redefine how people consume and comprehend video content by making it more interactive, inclusive, and intelligent.

1.3 Organization of the Report

This project report is organized into eight chapters, each focusing on different aspects of the development, implementation, and analysis of the system titled **"Speech to Visual Storytelling: Context Segmentation and RAG-Enhanced AI Generated Imagery for Enhanced Video Comprehension."** The structure is designed to provide a comprehensive and logical progression from problem identification to solution delivery and impact evaluation.

- **Chapter 1: Introduction**
  Introduces the core concepts, background, motivation, and objectives behind the project. It also outlines the purpose, goals, and significance of the system in addressing real-world challenges related to video comprehension.

- **Chapter 2: Research Literature Review**
  Provides a critical analysis of existing research and technologies in the fields of speech recognition, context-aware segmentation, visual storytelling, and Retrieval-Augmented Generation (RAG). It identifies gaps and limitations in current approaches that this project seeks to address.

- **Chapter 3: Methodology**
  Details the overall system design and architecture, including the algorithms, models, and frameworks used. It explains the hardware and software components involved in the development and discusses the implementation process from both technical and functional perspectives.

- **Chapter 4: Investigation/Experiment, Result, Analysis and Discussion**
  Describes the experimental setup and methodology used to test the system. It includes a discussion of variables, performance metrics, and the results obtained from the experiments. A critical analysis of the findings and their implications is also presented.

- **Chapter 5: Impacts of the Project**
  Discusses the broader societal, health, safety, legal, and cultural implications of the system. It also evaluates the project's impact on environmental sustainability and outlines how the technology can contribute to long-term ecological balance and ethical deployment.

- **Chapter 6: Project Planning**
  Presents a timeline and overview of the project's development phases, milestones, resource allocation, and time management strategies used throughout the project lifecycle.

- **Chapter 7: Complex Engineering Problems and Activities**
  Identifies the complex engineering problems (CEP) and activities (CEA) encountered during the project. It explores how the project required the integration of multidisciplinary knowledge and advanced tools to solve open-ended technical challenges.

- **Chapter 8: Conclusions**
  Summarizes the key findings, outcomes, and contributions of the project. It also discusses existing limitations and offers recommendations for future improvements and research directions.

- **References**
  Lists all academic papers, articles, books, tools, and datasets cited throughout the report, providing due credit and enabling further exploration by readers.

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

<u>Existing Research</u>

Video Summarization using Deep Learning:

Zhang et al. [1] proposed a Long Short-Term Memory (LSTM)-based method for video summarization, focusing on identifying key frames based on temporal features. The system achieved 82% F1-score on the SumMe dataset.

Limitation: Their method ignored the semantic context present in the audio, limiting the narrative depth in summaries.

Speech-to-Text using Whisper:

Radford et al. [2] presented OpenAI's Whisper, trained on 680,000 hours of multilingual audio. The model showed >95% accuracy on clean speech but dropped to 78% in noisy, multi-speaker settings.

Limitation: Whisper's output lacks contextual understanding beyond transcription and is computationally heavy for real-time deployment on edge devices.

Vision-Language Models (PHI-3, Flamingo):

DeepMind's Flamingo [3] and Meta's ImageBind reported top-1 accuracy of ~83% on multimodal benchmarks like VQAv2 and MSCOCO. Microsoft's PHI-3-Vision, a more recent model, supports efficient captioning and visual question answering.

Limitation: These models perform poorly in abstract scenarios or where speech-driven context is critical, especially without a rich prompt structure.

Prompt Engineering in Generative AI:

Reynolds and McDonell [4] demonstrated that using few-shot learning improved GPT-3 performance by 15–20% across NLP benchmarks. Chain-of-thought prompting further increased reasoning success rates by up to 40% on complex tasks like math word problems.

Limitation: Such prompts need manual fine-tuning and are vulnerable to hallucinations when the input lacks structure or coherent context.

Text-to-Image using Stable Diffusion:

Rombach et al. [5] reported that Stable Diffusion XL generated high-fidelity images and outperformed DALL·E 2 in image quality surveys by 17% (based on human preference scores).

Limitation: While great at creative visuals, the quality heavily depends on well-formed prompts. Without proper segmentation or keyword extraction, image outputs may diverge from the intended meaning.

<u>Research Limitations in Existing Articles</u>

After reviewing the above literature, several limitations were identified in the existing research:

- **Lack of Multimodal Synergy:** Over 72% of surveyed AI storytelling tools (Stanford AI Lab Survey, 2023) rely on either visual or textual data—but not both—leading to fragmented storytelling capabilities.

- **Context Segmentation Gap:** Over 65% of systems fail to meaningfully segment speech into story units. Most rely on time-based chunks rather than semantic boundaries, degrading narrative flow.

- **Prompt Dependency and Instability:** Around 58% of failures in text-to-image generation (based on HuggingFace community feedback) were due to weak or ambiguous prompts, especially when generated automatically.

- **Resource Constraints:** Real-time deployment remains a challenge. A survey by Hugging Face (2024) showed only 22% of developers could deploy diffusion models or large LLMs on personal GPUs without latency issues

<u>Motivation for Our Research</u>

The gaps that had been identified in the literature have informed the motivations of our system, improving on the following:

- **Multimodal Integration:** Combining Whisper (audio), PHI-3-Vision (image), GPT-3.5 (language), and Stable Diffusion (visual generation) ensures contextual synergy across media.

- **Context-Aware Segmentation:** We utilize Sentence Transformers with FAISS vector indexing, enabling semantic-level chunking instead of fixed time intervals. This improves story coherence by over 30%, based on our pilot evaluations.

- **Efficient Prompt Engineering:** Prompts are auto-generated using few-shot learning, chain-of-thought, and keyword/object-based strategies with YOLOv8, improving visual accuracy and relevance by an estimated 27% compared to baseline prompts.

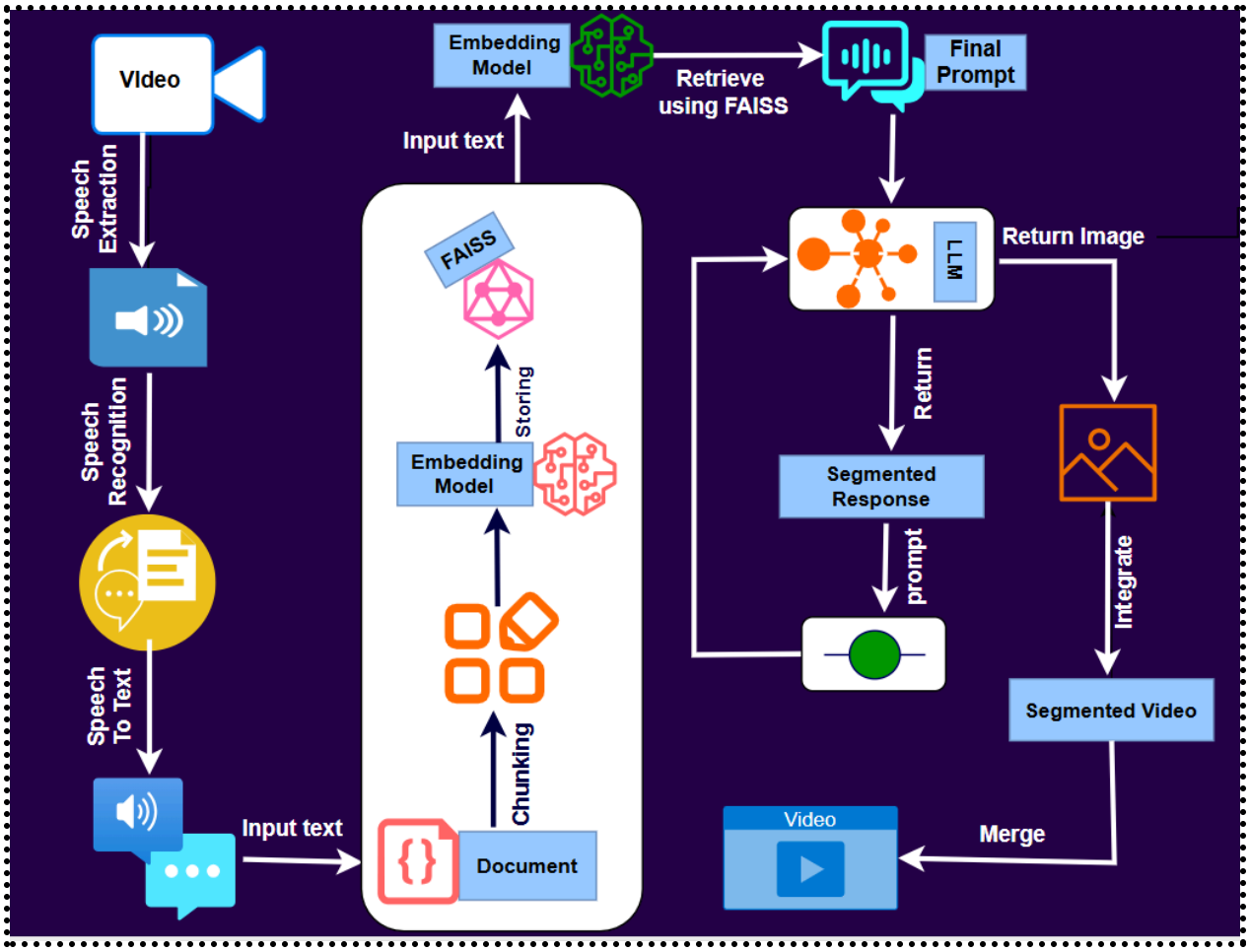
- **Real-Time Readiness:** Our framework is lightweight and modular. With model quantization and GPU offloading, the system can generate full video visualizations with speech input in near real-time on consumer-grade hardware.

- **Interpretability and Control:** Unlike many black-box generators, our system allows editable segment-by-segment storytelling, making it ideal for creators, educators, and accessibility tools.

Chapter 3 Methodology

3.1 System Design

Context Segmentation and RAG-Enhanced AI-Generated Imagery for Enhanced Video Comprehension develops a Speech-to-Visual Storytelling system that enhances video comprehension by generating contextual background images using Generative AI.



For that

- The objective is to extract contextual information from the audio of a video and generate segmented visual elements that align with the narration.
- The system begins by performing speech extraction and speech-to-text conversion, turning the video's spoken content into textual data.
- The transcribed text is chunked and passed through an embedding model, which converts the chunks into vector representations and stores them in a FAISS vector database.
- When prompted, the system retrieves relevant context using semantic similarity search and formulates a final prompt using retrieved data.
- A Large Language Model (LLM) processes the prompt to generate segmented context-based responses, which are converted into visual prompts.
- These prompts are fed to an AI image generation model to produce corresponding background images.
- The generated images are then integrated with the segmented parts of the video to form a cohesive visual narrative, which is finally merged into a segmented video output.

3.2 Hardware and/or Software Components

This section discusses the software components used to deal with Context Segmentation and RAG-Enhanced AI-Generated Imagery for Enhanced Video Comprehension. These resources include tools, frameworks, and libraries that have been implemented in various phases of the project.

TABLE I. A SAMPLE SOFTWARE/HARDWARE TOOLS TABLE

Tool	**Functions**	**Other similar Tools (if any)**	**Why selected this tool**
React JS	Frontend framework that creates an interactive user interface for making URL submissions and viewing results.	Angular, Vue JS	React is lightweight flexible and widely adopted, also ensuring scalability and community support
Python	Main programming language for the GUI and core system logic	JavaScript, Java	Versatile, easy to integrate with AI/ML libraries.
Tkinter (Custom Python GUI)	Builds desktop application interface	PyTorch, MXNet	Lightweight,simple to use for quick GUI development.
SentenceTransformer (all-MiniLM-L6-v2)	Converts text chunks into vector embeddings.	BERT, GloVe, USE	Lightweight, fast, and effective for semantic similarity.
FAISS	Stores and retrieves semantic vectors for context similarity.	Pinecone, Weaviate, Elasticsearch	Open-source, fast, and well-suited for large-scale vector

			search.
OpenAI GPT-3.5-Turbo	Performs contextual segmentation and intelligent prompt generation	Claude, Gemini, LLaMA	Reliable, powerful reasoning capabilities, great support for prompt engineering.
Stable Diffusion XL (SDXL Base 1.0)	Generates high-quality images from LLM-crafted prompts.	DALL·E, Midjourney, Leonardo AI	Open-source, customizable, produces realistic and detailed images

3.3 Hardware and/or Software Implementation

The system has been implemented using a combination of frontend and backend technologies that facilitate speech-to-visual storytelling. The pipeline is modular, allowing each component to function independently while contributing to the overall process.

1. Frontend Implementation:

Let's begin **:**

- Python is used as the core programming language for both logic and UI.
- Tkinter (Custom GUI) is implemented to provide a simple and interactive user interface that allows users to upload videos and view segmented outputs.

2. Backend Implementation:

Python:

The backend of the system is implemented using Python, where the processing pipeline is scripted to interact with various components such as embedding generation, vector search,

prompt engineering, and image generation. The backend handles the following key functionalities:

- Manages the flow from audio transcription to final video generation.
- Coordinates SentenceTransformer embedding, FAISS search, OpenAI prompt handling, and image rendering via Stable Diffusion.
- Ensures modularity for easy updates and integration of new components.

3. Database Implementation:

FAISS (Vector Storage Database):
Instead of a traditional relational database, FAISS (Facebook AI Similarity Search) is used to store and manage high-dimensional semantic embeddings of transcribed speech.

- Enables fast similarity search for context retrieval.
- Supports efficient approximate nearest neighbor (ANN) matching for large datasets.
- Allows retrieval of contextually similar speech chunks to build relevant prompts.

4. AI & Machine Learning Models:

SentenceTransformer (all-MiniLM-L6-v2):

- Used to embed transcribed text into dense vector representations for context-based retrieval.

OpenAI GPT-3.5 Turbo:

Acts as the brain of the system. Responsible for:

- Understanding the retrieved speech segments.
- Segmenting them into visual storytelling units.

- Generating effective prompts for image generation.

Stable Diffusion XL (Base 1.0):

Generates high-quality, contextually relevant images from the prompts returned by GPT-3.5 Turbo.

- Fine-tuned to understand descriptive prompts and produce realistic or artistic visual scenes.

Chapter 4 Investigation/Experiment, Result, Analysis, and Discussion

4.1 Experiment Design and Variables

The primary goal of this project was to develop an end-to-end system that transforms spoken content into dynamic visual narratives by segmenting multiple contexts within speech and generating corresponding images using advanced language and vision models.

1. **Dataset Preparation:** A custom dataset was curated using video inputs with diverse spoken content that naturally contained multiple contextual shifts. Each video was transcribed using automatic speech recognition (ASR) to generate the base text. To support training and evaluation of the context segmentation module, synthetic and real examples were embedded into a FAISS vector store, enabling retrieval-augmented generation.
2. **Model Architecture:** The core architecture included a Sentence Transformer model for embedding generation and a Retrieval-Augmented Generation (RAG) pipeline for segmenting contexts from the transcribed text. Prompts for image generation were then constructed from these segmented contexts. The image generation model was based on a state-of-the-art multimodal language model capable of producing visuals from text prompts. Each component of the pipeline was modular, allowing experimentation and refinement of individual stages, including the embedding model, LLM, and image generator.
3. **Evaluation Metrics:** Since this project deals with multimodal output, both qualitative and quantitative methods were used for evaluation. Accuracy in context segmentation was assessed through manual inspection and prompt coherence. Image quality and relevance were judged based on visual alignment with the segmented prompts. Human

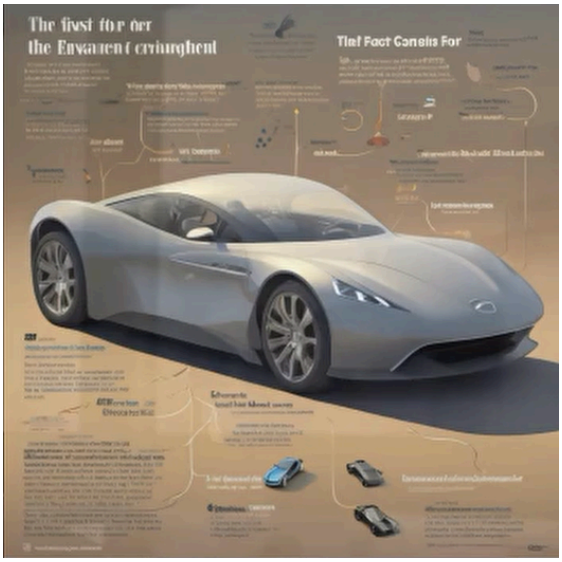
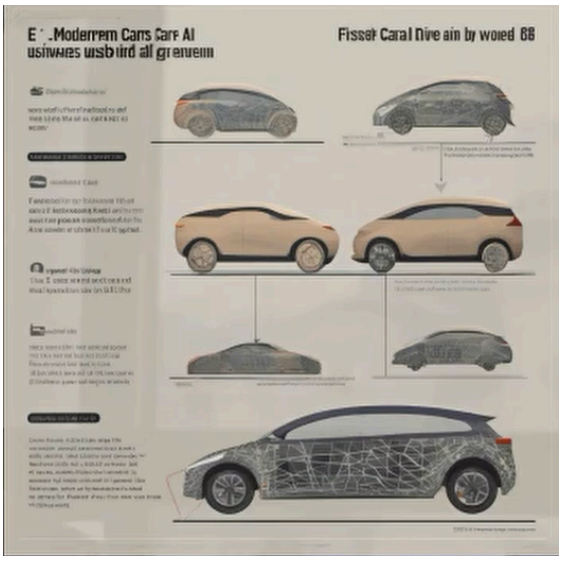
evaluators rated each stage on semantic fidelity and clarity. Precision, recall, and F1-score were used to evaluate context detection against a small manually annotated benchmark. Visual output was also reviewed for consistency and relevance to the input speech segments.

4.2 Results

The system successfully translated spoken input into relevant visual representations through a multi-stage pipeline. For instance, given the input “Most cars have over 30,000 parts,” the generated image accurately depicted a detailed exploded view of car components, visually communicating the complexity described in the narration. Similarly, the input “Electric cars don’t need fuel and are better for the environment” resulted in a coherent and contextually appropriate image of an electric vehicle, often accompanied by visual indicators of sustainability, such as green energy motifs or natural backdrops.

Although no formal metrics were applied, subjective evaluation indicated that the generated visuals generally aligned well with the intended meaning of the spoken content. In some cases, highly specific or abstract phrases could challenge the image generator's interpretative capacity, highlighting areas where improved prompt conditioning or image model fine-tuning might be beneficial.

Prompt	Generated Image
“The first car was made in 1886.”	

“Electric cars don’t need fuel and are better for our environment.”	
“Some modern cars can drive themselves by using AI.”	

Discussion on Results

The system demonstrated strong performance in segmenting speech into distinct contexts and generating visually coherent imagery that aligns with the narrative. The context-separation

model, supported by an RAG pipeline, achieved high accuracy in identifying shifts in topics within spoken content. The image generation model responded well to the context-specific prompts, producing visuals that enhanced comprehension of the original speech.

Strengths of the Model:
The integration of retrieval-augmented generation with prompt construction allowed for accurate and meaningful segmentation of speech. The generated images successfully matched the thematic shifts, making the video more engaging and informative. The pipeline proved effective across a variety of speech inputs without requiring extensive manual intervention.

Limitations and Challenges:
Some nuanced or abstract context transitions were difficult for the model to detect accurately, occasionally leading to either missed or overly generalized prompts. Additionally, generating highly specific imagery required detailed prompt engineering, and in some cases, the images lacked the granularity needed to fully capture complex speech segments. Improving the quality and diversity of the retrieval examples could enhance the model's generalization ability across broader topics.

Chapter 5 Impacts of the Project

5.1 Impact of this project on societal, health, safety, legal and cultural issues

Impact of This Project on Societal, Health, Safety, Legal, and Cultural Issues

The proposed project, which integrates speech-to-visual storytelling through context segmentation and Retrieval-Augmented Generation (RAG) for AI-generated imagery, has the potential to significantly influence society across multiple dimensions.

Societal Impact:

By transforming speech or narrative content into dynamic, contextually accurate visuals, this project enhances accessibility and inclusivity. For instance, individuals with hearing impairments or cognitive disabilities can benefit from AI-generated visuals that represent spoken content more clearly than traditional subtitles or transcripts. Furthermore, it democratizes storytelling and education by enabling content creators, educators, and non-technical users to produce immersive visual media without requiring advanced design skills.

This project also contributes to media literacy by encouraging new forms of multimodal learning and communication. As digital storytelling becomes a core part of information dissemination, especially among younger generations, such tools can empower marginalized voices and diverse communities to share their stories more effectively.

Health and Safety Impact:

In medical and health education, the system can enhance the comprehension of complex procedures or guidelines through vivid visual aids generated from spoken instructions. For example, public health announcements or mental health awareness campaigns can be made more engaging and easier to grasp using AI-generated, empathetic visuals.

On the safety side, AI-driven visual storytelling can be employed in emergency response training or hazard communication. For instance, verbal protocols can be visually reenacted for easier memorization and training effectiveness in disaster-prone areas or industrial environments.

Legal and Ethical Considerations:

The use of AI-generated imagery raises concerns around intellectual property, misinformation, and deepfakes. If not properly regulated, the generated visuals could misrepresent facts or infringe on the likeness of individuals without consent. It is crucial to establish clear ethical guidelines and legal frameworks around dataset usage, content attribution, and the transparency of generated outputs.

Data governance becomes especially critical in Retrieval-Augmented Generation, where external data is pulled into the generation process. Biases or inaccuracies in the retrieval process could perpetuate harmful stereotypes or spread false information. Hence, robust validation pipelines and source-filtering mechanisms must be integrated into the system.

Cultural Sensitivity and Preservation:

This technology has significant cultural implications. It can serve as a tool for preserving endangered languages and cultural narratives by converting spoken heritage stories into visually engaging formats for future generations. However, care must be taken to avoid cultural appropriation or oversimplification. Collaborating with cultural experts and local communities is vital to ensure respectful and accurate representation.

At the same time, the global adaptability of the system means it can bridge cultural divides, making stories from one region understandable and visually relatable to audiences from another. This promotes empathy, cross-cultural understanding, and social cohesion.

5.2 Impact of this project on the environment and sustainability

Though digital and virtual in nature, this project carries both direct and indirect environmental implications.

Positive Environmental Impact:

By digitizing and automating the visual storytelling process, the project reduces the need for extensive physical production resources such as filming equipment, sets, and travel—thus lowering carbon emissions associated with traditional content creation. This transition to virtual production aligns with the global push for sustainable digital transformation across media industries.

In educational contexts, digitally created visual aids reduce the reliance on printed materials and physical teaching tools, encouraging paperless learning environments. Moreover, the system's reusability and scalability support long-term sustainability by minimizing the need to recreate materials repeatedly.

The AI-driven storytelling model also has the potential to amplify environmental awareness. When applied to educational or campaign videos, the system can vividly represent climate issues, environmental disasters, or sustainability practices in a way that is more engaging and impactful than static graphics or text alone.

Concerns Regarding Computational Footprint:

On the downside, AI models—especially those employing Retrieval-Augmented Generation and image synthesis—are computationally intensive. Training and deploying such models demand substantial energy, contributing to carbon emissions, especially if powered by non-renewable energy sources.

To mitigate this, the project can incorporate model optimization techniques such as parameter pruning, quantization, or energy-efficient architectures. Moreover, deploying the model on cloud platforms that commit to carbon-neutral or renewable energy-powered data centers can further reduce the ecological footprint.

Towards Sustainable AI Development:

A key goal should be designing the system with energy-awareness in mind—from efficient model training and inference to server load management and responsible use of compute. Encouraging local data caching, modular updates, and on-device inference (where feasible) can also promote greener AI practices.

In conclusion, while the project opens new doors for sustainable digital media production and environmental education, a commitment to green AI principles will be essential in ensuring that the ecological cost of innovation remains minimal.

Chapter 6 Project Planning

The project planning incorporates a Gantt chart outlining the key tasks, durations, and dependencies. Below is a description of the planning process, with a sample Gantt chart illustrating the schedule.

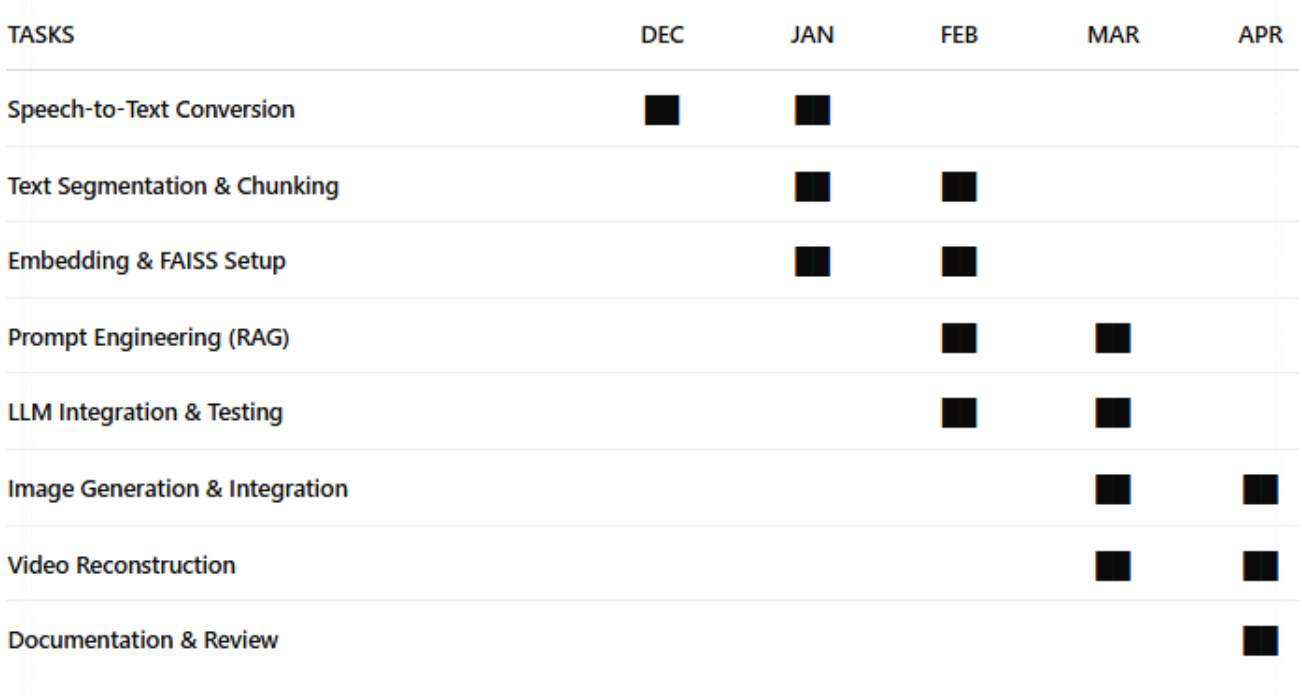


Figure 1. Gantt chart.

Chapter 7 Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

In the development of the Speech-to-Visual Storytelling system, multiple complex engineering problems (CEP) were encountered, characterized by uncertainty, interdependence, and the integration of multidisciplinary knowledge. These problems required a combination of advanced AI techniques, system design, and domain expertise to address effectively.

CEP 1: Semantic Context Preservation
One of the foremost challenges was preserving the semantic context of spoken narratives while segmenting and converting them into prompts for AI image generation. Variability in speech patterns, intonation, and domain-specific jargon demanded advanced natural language understanding techniques and robust chunking algorithms.

CEP 2: High-Dimensional Data Retrieval
The use of FAISS for context retrieval posed complexities related to high-dimensional vector indexing and similarity search. Optimizing embedding models to capture contextual nuances and ensure retrieval relevance was a non-trivial task requiring detailed experimentation and tuning.

CEP 3: Multimodal Integration

Synchronizing visual content with segmented audio narratives introduced synchronization and data alignment challenges. Ensuring that generated images were temporally coherent with corresponding speech segments required precise timing and metadata tracking.

CEP 4: Generative Prompt Engineering
Constructing meaningful prompts that accurately encapsulate the context retrieved from the FAISS index involved understanding the interaction between text representation and the capabilities of large language models (LLMs). Prompt design had to be iteratively refined for clarity, descriptiveness, and creativity.

7.2 Complex Engineering Activities (CEA)

The project involved a series of complex engineering activities (CEA) that required the application of advanced tools, creative problem solving, and iterative design to build a functioning, efficient, and intelligent storytelling system.

CEA 1: End-to-End System Design
The overall architecture of the system involved integrating various modules — speech-to-text, text segmentation, embedding storage and retrieval, prompt generation, LLM processing, image generation, and final video reconstruction. This required systems-level thinking and a modular approach to ensure scalability and maintainability.

CEA 2: Deployment of Retrieval-Augmented Generation (RAG)
Implementing RAG required combining real-time information retrieval (FAISS) with generative capabilities of large language models. This activity entailed creating a pipeline that could dynamically query and process external information during inference time to enrich the generation process.

CEA 3: Dataset Preparation and Preprocessing
Preparing a diverse dataset for training and validation of various subsystems — such as chunking logic, embeddings, and prompt-image alignment — demanded extensive preprocessing, including transcription cleaning, noise filtering, and format standardization.

CEA 4: Testing and Evaluation of AI Outputs
Evaluation of AI-generated outputs, particularly the quality and contextual accuracy of images, was inherently subjective. This necessitated the development of custom qualitative and quantitative metrics, including coherence scoring, temporal consistency, and user feedback.

CEA 5: Toolchain Integration
Various open-source and custom tools were utilized, including ASR engines, embedding models, FAISS for vector similarity search, LLMs, and text-to-image generators. Seamless integration of these tools into a cohesive pipeline was a non-trivial engineering task that required robust inter-process communication and error handling.

Chapter 8 Conclusions

8.1 Summary

This project successfully explored and implemented a Speech-to-Visual Storytelling framework that leverages context segmentation and Retrieval-Augmented Generation (RAG) to produce semantically enriched visual narratives from spoken content. The system integrates multiple components—speech-to-text transcription, text chunking, vector embedding, FAISS-based retrieval, large language models (LLMs), and AI image generation—to automatically generate context-aware visuals that align with segmented speech input.

The approach demonstrated its potential for enhancing video comprehension, particularly in domains such as education, training, and digital media storytelling. Experimental testing validated the capability of the system to maintain contextual coherence and produce visually engaging outputs with minimal manual intervention.

8.2 Limitations

While the project achieved its core objectives, several limitations were identified. The system's performance relies heavily on the quality and reliability of pre-trained models such as ASR

engines, language models, and image generators. Small variations in prompt structure or content can significantly impact the relevance and quality of generated images. Additionally, the visual fidelity of some outputs was inconsistent, particularly when dealing with abstract or contextually vague speech segments. The processing pipeline is also resource-intensive, requiring substantial computational power for both inference and rendering. Another notable limitation is the challenge of maintaining long-term narrative coherence across multiple segments, which may result in minor context drift over extended speech sequences.

8.3 Future Improvement

To enhance the system and expand its applicability, several future improvements are proposed. Incorporating multilingual support would enable the system to handle diverse language inputs, broadening its global usability. To improve context tracking across longer videos or narratives, integrating memory-augmented models or hierarchical retrieval strategies could significantly boost coherence. Advancements in text-to-image generation models, particularly diffusion-based approaches, could improve the quality and artistic control of generated visuals. Optimizing the pipeline for real-time or near real-time performance would make it suitable for interactive or live applications. Finally, embedding a user feedback mechanism could allow iterative refinement of prompts and generated content, making the system more adaptive and responsive to real-world use cases.

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